Chapter 1

Bayesian Linear Regression

§ 1.1

Basic Settings

Consider a training dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N},$$

consisting of N observations, where $\boldsymbol{x}_i \in \mathbb{R}^p$ denotes the p-dimensional feature vector associated with the i-th observation and $y_i \in \mathbb{R}$ denotes the corresponding scalar response.

Define design matrix $\mathbf{X}$, target vector $\boldsymbol{y}$, and coefficient vector $\boldsymbol{w}$ as

$$\mathbf{X} = \begin{bmatrix} \boldsymbol{x}_1^\top \\ \vdots \\ \boldsymbol{x}_N^\top \end{bmatrix} \in \mathbb{R}^{N\times p}, \qquad \boldsymbol{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_N \end{bmatrix} \in \mathbb{R}^{N}, \qquad \boldsymbol{w} = \begin{bmatrix} w_1 \\ \vdots \\ w_p \end{bmatrix} \in \mathbb{R}^{p}.$$

Under the linear regression model, each observation satisfies

$$y_i = \boldsymbol{w}^\top \boldsymbol{x}_i + \epsilon_i,$$

where the noise terms are assumed to be independent and identically distributed as

$$\epsilon_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2), \qquad i = 1, \dots, N.$$

Define the noise vector

$$\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_N \end{bmatrix}.$$

It follows that

$$\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N),$$

where $\mathbf{I}_N$ denotes the $N \times N$ identity matrix.

For the entire dataset, the linear regression model can therefore be written as

$$\boldsymbol{y} = \mathbf{X}\boldsymbol{w} + \boldsymbol{\epsilon}.$$

Consequently,

$$p(\boldsymbol{y}|\mathbf{X}, \boldsymbol{w}) \sim \mathcal{N}(\mathbf{X}\boldsymbol{w}, \sigma^2\mathbf{I}_N).$$

Bayesian method includes two basic steps: inference and prediction, where inference is to estimate posterior $\boldsymbol{w}$, while prediction is to predict y^* given $\boldsymbol{x}^*$.

§ 1.2 Inference

1.2.1 Model Establishment

In Bayesian linear regression, the coefficient vector $\boldsymbol{w}$ is treated as a random variable and assigned a prior distribution. We assume that the prior distribution of $\boldsymbol{w}$ does not depend on $\mathbf{X}$, i.e., $w \perp X$, and thus $p(\boldsymbol{w} \mid \mathbf{X}) = p(\boldsymbol{w})$.

Given the training dataset, Bayesian formula is decomposed as:

$$p(\boldsymbol{w}|\mathcal{D}) = p(\boldsymbol{w}|\mathbf{X}, \boldsymbol{y}) = \frac{p(\boldsymbol{w}, \mathbf{X}, \boldsymbol{y})}{p(\mathbf{X}, \boldsymbol{y})} = \frac{p(\boldsymbol{w}, \mathbf{X}, \boldsymbol{y})}{p(\boldsymbol{y}|\mathbf{X})p(\mathbf{X})} = \frac{p(\boldsymbol{w}, \boldsymbol{y}|\mathbf{X})}{p(\boldsymbol{y}|\mathbf{X})} = \frac{p(\boldsymbol{y}|\mathbf{X}, \boldsymbol{w})p(\boldsymbol{w}|\mathbf{X})}{p(\boldsymbol{y}|\mathbf{X})} = \frac{p(\boldsymbol{y}|\mathbf{X}, \boldsymbol{w})p(\boldsymbol{w})}{p(\boldsymbol{y}|\mathbf{X})},$$

where the fifth equality follows from

$$p(\boldsymbol{y}|\mathbf{X}, \boldsymbol{w}) = \frac{p(\boldsymbol{w}, \mathbf{X}, \boldsymbol{y})}{p(\boldsymbol{w}, \mathbf{X})} = \frac{p(\boldsymbol{w}, \mathbf{X}, \boldsymbol{y})}{p(\boldsymbol{w}|\mathbf{X})p(\mathbf{X})} = \frac{p(\boldsymbol{w}, \boldsymbol{y}|\mathbf{X})}{p(\boldsymbol{w}|\mathbf{X})},$$

and the last equality follows from independency of prior coefficient and design matrix.

The denominator

$$p(\boldsymbol{y} \mid \mathbf{X}) = \int p(\boldsymbol{y} \mid \mathbf{X}, \boldsymbol{w})p(\boldsymbol{w})\mathrm{d}\boldsymbol{w}$$

is the marginal likelihood. Since it does not depend on $\boldsymbol{w}$, the posterior distribution can be expressed up to a proportionality constant as

$$p(\boldsymbol{w}|\mathbf{X}, \boldsymbol{y}) \propto p(\boldsymbol{y}|\mathbf{X}, \boldsymbol{w})p(\boldsymbol{w}),$$

where $p(\boldsymbol{y}|\mathbf{X}, \boldsymbol{w})$ is a likelihood function and $p(\boldsymbol{w})$ is a prior function.

Suppose that a zero-mean Gaussian prior is imposed on the coefficient vector:

$$p(\boldsymbol{w}) = \mathcal{N}\left(\mathbf{0}, \boldsymbol{\Sigma}_p\right),$$

where $\mathbf{\Sigma}_p \in \mathbb{R}^{p\times p}$ denotes the prior covariance matrix.

Since both the likelihood and the prior are Gaussian, their conjugacy implies that the posterior distribution of $\boldsymbol{w}$ is also Gaussian:

$$p(\boldsymbol{w}|\mathbf{X},\boldsymbol{y}) = \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{w}},\mathbf{\Sigma}_{\boldsymbol{w}})$$

Thus,

$$p(\boldsymbol{w}|\mathbf{X},\boldsymbol{y}) \propto \mathcal{N}(\mathbf{X}\boldsymbol{w},\sigma^2\mathbf{I}_N)\mathcal{N}\left(\mathbf{0},\mathbf{\Sigma}_p\right).$$

The remaining task is therefore to derive the posterior mean $\boldsymbol{\mu}_w$ and posterior covariance $\mathbf{\Sigma}_w$.

1.2.2 Model Solution

To derive the posterior distribution, we first write the product of the likelihood and the prior explicitly:

$$\mathcal{N}(\mathbf{X}\boldsymbol{w},\sigma\mathbf{I}_N)\mathcal{N}\left(\mathbf{0},\mathbf{\Sigma}_p\right) = \frac{1}{(2\pi)^{N/2}\sigma^N}\exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{y}-\mathbf{X}\boldsymbol{w})^\top(\boldsymbol{y}-\mathbf{X}\boldsymbol{w})\right)\frac{1}{(2\pi)^{p/2}|\mathbf{\Sigma}_p|^{1/2}}\exp\left(-\frac{1}{2}(\boldsymbol{w})^\top\mathbf{\Sigma}_p^{-1}\boldsymbol{w}\right).$$

Thus,

$$\begin{aligned}
p(\boldsymbol{w}|\mathbf{X},\boldsymbol{y}) &\propto \exp\left(-\frac{1}{2\sigma^2}(\boldsymbol{y}-\mathbf{X}\boldsymbol{w})^\top(\boldsymbol{y}-\mathbf{X}\boldsymbol{w})-\frac{1}{2}(\boldsymbol{w})^\top\mathbf{\Sigma}_p^{-1}\boldsymbol{w}\right)\\
&= \exp\left(-\frac{1}{2\sigma^2}\left(\boldsymbol{y}^\top\boldsymbol{y}-\boldsymbol{y}^\top\mathbf{X}\boldsymbol{w}-\boldsymbol{w}^\top\mathbf{X}^\top\boldsymbol{y}+\boldsymbol{w}^\top\mathbf{X}^\top\mathbf{X}\boldsymbol{w}\right)-\frac{1}{2}(\boldsymbol{w})^\top\mathbf{\Sigma}_p^{-1}\boldsymbol{w}\right)\\
&= \exp\left(-\frac{1}{2\sigma^2}\boldsymbol{y}^\top\boldsymbol{y}+\frac{1}{\sigma^2}\boldsymbol{w}^\top\mathbf{X}^\top\boldsymbol{y}-\frac{1}{2}\boldsymbol{w}^\top\left(\frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}+\mathbf{\Sigma}_p^{-1}\right)\boldsymbol{w}\right).
\end{aligned}$$

Comparing this expression with the kernel of a multivariate Gaussian distribution,

$$\exp\left(-\frac{1}{2}(\boldsymbol{w}-\boldsymbol{\mu}_{\boldsymbol{w}})^\top\mathbf{\Sigma}_{\boldsymbol{w}}^{-1}(\boldsymbol{w}-\boldsymbol{\mu}_{\boldsymbol{w}})\right) = \exp\left(-\frac{1}{2}\boldsymbol{w}^\top\mathbf{\Sigma}_{\boldsymbol{w}}^{-1}\boldsymbol{w}+\boldsymbol{w}^\top\mathbf{\Sigma}_{\boldsymbol{w}}^{-1}\boldsymbol{\mu}_{\boldsymbol{w}}-\frac{1}{2}\boldsymbol{\mu}_{\boldsymbol{w}}^\top\mathbf{\Sigma}_{\boldsymbol{w}}^{-1}\boldsymbol{\mu}_{\boldsymbol{w}}\right),$$

we obtain

$$\mathbf{\Sigma}_{\boldsymbol{w}}^{-1} = \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X}+\mathbf{\Sigma}_p^{-1} = \mathbf{A}.$$

Matching the linear terms gives

$$\boldsymbol{w}^\top\mathbf{\Sigma}_{\boldsymbol{w}}^{-1}\boldsymbol{\mu}_{\boldsymbol{w}} = \frac{1}{\sigma^2}\boldsymbol{w}^\top\mathbf{X}^\top\boldsymbol{y} \Leftrightarrow \mathbf{A}\boldsymbol{\mu}_{\boldsymbol{w}} = \frac{1}{\sigma^2}\mathbf{X}^\top\boldsymbol{y},$$

and therefore

$$\boldsymbol{\mu}_{\boldsymbol{w}} = \frac{1}{\sigma^2}\mathbf{A}^{-1}\mathbf{X}^\top\boldsymbol{y}.$$

Hence, the posterior distribution has the closed-form solution

$$p(\boldsymbol{w}|\mathbf{X},\boldsymbol{y}) \sim \mathcal{N}(\boldsymbol{\mu}_{\boldsymbol{w}},\mathbf{\Sigma}_{\boldsymbol{w}}),$$

where

$$\mathbf{\Sigma}_{\boldsymbol{w}}^{-1} = \frac{1}{\sigma^2}\mathbf{X}^\top\mathbf{X} + \mathbf{\Sigma}_p^{-1}, \qquad \boldsymbol{\mu}_{\boldsymbol{w}} = \frac{1}{\sigma^2}\mathbf{\Sigma}_{\boldsymbol{w}}\mathbf{X}^\top\boldsymbol{y}.$$

§ 1.3 Prediction

After obtaining the posterior distribution of the coefficient vector, Bayesian linear regression can be used to predict the responses associated with new observations.

Let $\mathbf{X}^* \in \mathbb{R}^{N^* \times p}$ denote the design matrix for N^* new observations, and let $\boldsymbol{y}^* \in \mathbb{R}^{N^*}$ denote the corresponding target vector to be predicted.

Given the posterior distribution, the posterior predictive distribution is obtained by marginalizing over the uncertainty in $\boldsymbol{w}$:

$$p(\boldsymbol{y}^* \mid \mathbf{X}^*, \mathcal{D}) = \int p(\boldsymbol{y}^* \mid \mathbf{X}^*, \boldsymbol{w})p(\boldsymbol{w} \mid \mathcal{D})\mathrm{d}\boldsymbol{w}.$$

Thus, we have

$$p(\boldsymbol{y}^* \mid \mathbf{X}^*, \mathcal{D}) = \mathcal{N}\left(\mathbf{X}^*\boldsymbol{\mu}_{\boldsymbol{w}}, \mathbf{X}^*\mathbf{\Sigma}_w(\mathbf{X}^*)^\top + \sigma^2\mathbf{I}_{N^*}\right).$$

Therefore, the predictive mean and covariance are

$$\mathbb{E}\left[\boldsymbol{y}^* \mid \mathbf{X}^*, \mathcal{D}\right] = \mathbf{X}^*\boldsymbol{\mu}_w,$$

and

$$\mathrm{Cov}\left(\boldsymbol{y}^* \mid \mathbf{X}^*, \mathcal{D}\right) = \underbrace{\mathbf{X}^*\mathbf{\Sigma}_w(\mathbf{X}^*)^\top}_{\text{parameter uncertainty}} + \underbrace{\sigma^2\mathbf{I}_{N^*}}_{\text{observation noise}}.$$